

Exploring a Regulatory Mapping Approach for Designing Digital Mental Health Interventions within the EU Context

Kang, Heimin; Jung, Jiwon

As digital mental health interventions, such as generative AI chatbots, expand, integrating regulatory guidelines into the design process remains essential but challenging due to their dynamic and complex nature. This study examines the impact of six key EU regulations on the design of AI chatbots as representative digital mental health interventions. By mapping these regulations onto a framework for developing and evaluating complex health interventions, this study provides an overview of the regulatory landscape at each stage. Findings highlight the tension between regulation and innovation while identifying opportunities to integrate regulatory compliance into design methodologies. This research contributes to understanding and applying regulations in developing and evaluating complex health interventions, enabling designers to develop innovative yet compliant technologies. Limitations include a context-specific approach that limits generalizability and the evolving nature of regulations, which affects long-term applicability. Future work involves expert validation and iterative refinement to enhance accuracy and real-world applicability.

Keywords: *Digital mental health, Artificial intelligence, Design process innovation, EU regulations, Chatbot*

Expected Contributions

- Proposes a process-based regulatory mapping approach to support the development of digital mental health interventions, translating complex regulatory requirements into an understandable and actionable form for designers.
- Identifies tensions between innovation and regulatory compliance and offers design-led opportunities for integrating regulation into practice.
- Provides a methodological lens for embedding regulatory considerations into the development and evaluation of complex health interventions.